8.7. AQUEOUS SOLUTIONS

In irradiated dilute aqueous solutions at concentrations <0.1 mol/l, practically all the energy absorbed is deposited in the water molecules. Hence, the observed chemical changes are the result of the reactions between the solutes and the products of the water radiolysis. With increasing solute concentrations, the direct radiolysis of the solute gradually becomes important and the solute may also interfere with the spur reactions. The use of high concentrations of solutes which react selectively with the radical products — so-called *scavengers* — have provided experimental support for the existence of spurs.

In dilute solutions the chemical changes of a specific solute primarily reflect its reactivity towards e_{aq}^-, $H\cdot$ and $\cdot OH$. The hydrated electron e_{aq}^- is a strongly reducing species ($E_0 = -2.9$ V) whereas the hydrogen atom is a less powerful reductant ($E_0 = -2.3$ V). The H atom can be considered as a weak acid with a pKa of 9.6

$$e_{aq}^- + H^+ \rightleftharpoons H\cdot \tag{8.19}$$

The hydroxyl radical $\cdot OH$ is a strong oxidant ($E_0 = 2.7$ V in acidic and 1.8 V in basic solution). In strongly alkaline solution $\cdot OH$ dissociates into its anionic form O^- which has nucleophilic properties

$$\cdot OH \rightleftharpoons \cdot O^- + H^+ (pKa = 11.9) \tag{8.20}$$

In oxygenated solutions the perhydroxyl radical $\cdot HO_2$ is formed in reactions between e_{aq}^-, $\cdot H$ and O_2

$$e_{aq}^- + O_2 \rightarrow \cdot O_2^- \tag{8.21}$$

$$H\cdot + O_2 \rightarrow \cdot HO_2 \tag{8.22}$$

$$\cdot HO_2 \rightleftharpoons \cdot O_2^- + H^+ \ (pKa = 4.7) \tag{8.23}$$

In saline waters (0.1 M Cl^-, pH 4–10) the dominant radiolysis reactions are

$$\cdot OH + Cl^- \rightleftharpoons HClO^- \tag{8.24}$$

$$HClO^- + H^+ \rightleftharpoons Cl\cdot + H_2O \tag{8.25}$$

$$Cl^- + Cl\cdot \rightleftharpoons Cl_2^- \tag{8.26}$$

These equilibria are pH-dependent, and the reactions may have important consequences for storage of nuclear waste in salt media.

The oxidation of Fe^{2+} to Fe^{3+} in acidic oxygenated solution is the basis for the *Fricke dosimeter*, §8.10. The reaction scheme is:

$$e_{aq}^- + H^+ \rightarrow H\cdot \tag{8.27}$$

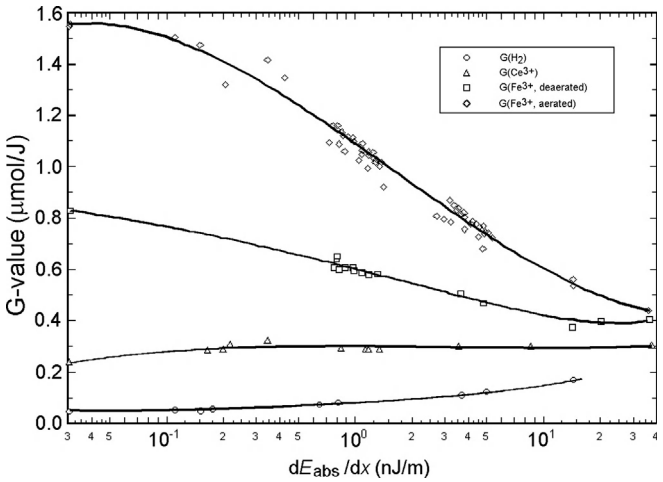


Figure 8.7 Radiation yield for oxidation of Fe^{2+} and reduction of Ce^{4+} in slightly acid (H_2SO_4) solution as a function of the radiation *LET* value. *(From A. O. Allen.)*

$$H\cdot + O_2 \rightarrow \cdot HO_2 \tag{8.28}$$

$$\cdot HO_2 + Fe^{2+} \rightarrow Fe^{3+} + HO_2^- \tag{8.29}$$

$$HO_2^- + H^+ \rightarrow H_2O_2 \tag{8.30}$$

$$H_2O_2 + Fe^{2+} \rightarrow Fe^{3+} + OH^- + \cdot OH \tag{8.31}$$

$$\cdot OH + Fe^{2+} \rightarrow Fe^{3+} + OH^- \tag{8.32}$$

The yield of Fe^{3+} is given by the equation

$$G\left(Fe^{3+}\right) = 2G(H_2O_2) + 3\left[G\left(e_{aq}^-\right) + G(\cdot H) + G(\cdot HO_2)\right] + G(\cdot OH) \tag{8.33}$$

Figure 8.7 shows radiation yields for oxidation of Fe^{2+} in acidic solution as a function of the *LET* value (see also §8.10).

As pointed out earlier, when the concentration of the aqueous solution is greater than about 0.1 M, the solutes may undergo direct radiolysis. The products of the radiolysis of the solute can react with water itself or with the radiolytic products of water. Irradiation of solutions containing sulfate, sulfite, or sulfide ions with *fast neutrons* yields radioactive phosphorus, through the $^{32}S(n,p)^{32}P$ reaction, almost exclusively as orthophosphate. However, depending on the redox conditions of the solution, reduced species of phosphorus also appear in minor amounts. Somewhat in contrast to this, *slow neutron* irradiation of solutions of $NaHPO_4$ yields phosphorus in many different oxidation states;

thus P(+1), P(+3) and P(+5) appear in species such as hypophosphite, phosphite, and o-phosphate. These species are all rather stable in aqueous solutions, and have been identified through paper electrophoretic analysis. In general, it may be assumed that slow neutron irradiation of solutions of oxyanions changes the central atom to another (usually a more reduced) valence state through n,y-reactions. For example, while manganese is in the Mn(+7) state in MnO_4^-, neutron capture leads to the formation of $^{56}Mn^{2+}$ species. However, if the product valence state is unstable at ambient solution conditions, it may be immediately oxidized to a more stable higher valence state.

The *self radiolysis* of a solution may change the chemical equilibria of the solution components. For example, the α-decay of plutonium decomposes water; in a solution containing 1 mole of ^{239}Pu, ca. 0.01 mole of H_2O_2 is produced per day. This hydrogen peroxide can react with the plutonium to form a precipitate of plutonium peroxide. To avoid this precipitation, nitrite ions are added to the solution to react with the hydroxyl radicals formed by the radiolysis and to eliminate the H_2O_2.

8.8. ORGANIC COMPOUNDS

As discussed above, the chemical consequences of radiolysis depends on the physical state and the molecular composition of the irradiated material. Two properties, the dielectric constant and the electron mobility, are of great importance for the fate of the ion pairs (the radical cations and the electrons) formed on ionization.

The distance r_c (the *Onsager radius*) at which the potential energy of an ion pair corresponds to the thermal energy $\mathbf{k}T$ is, according to Onsager,

$$r_c = \mathbf{e}^2/4\pi\,\boldsymbol{\varepsilon}_0\,\varepsilon_r\,\mathbf{k}\,T \tag{8.34}$$

where $\boldsymbol{\varepsilon}_0$ and ε_r is the permittivity of free space and in the medium at distance r_c, respectively (the *Onsager equation*). The probability of an electron escaping its positive ion to become a free ion is equal to $e^{-r_c/r}$ (r is the distance travelled by the electron before it becomes thermalized) depends on the electron mobility in the medium. Most organic liquids have a lower dielectric constant than that of water and the free ion yields are, therefore, generally lower, see Table 8.4.

Table 8.4 Onsager radii, electron mobilities and free ion yields

Liquid	r_c (nm)	Mobility (cm^2/Vs)	G(free ions) (μmol/J)
Neopentane	32	55	0.09–0.11
Cyclohexane	28	0.35	0.016–0.02
Benzene	25	-	0.005–0.008
Methanol	2.3	-	0.2
Water	0.7	-	0.28